

Credit Card Transaction Risk & Statistical Analysis System

Project Technical Documentation & Analysis Summary | GitHub Repository Reference

Project Overview: This project constructs a synthetic credit card transaction dataset comprising 1,000 records to perform exploratory data analysis, evaluate risk status distributions, and analyze fraud detection attributes across transaction hours, types, and locations.

1. Dataset Generation & Schema Setup

The synthetic dataset uses numpy's lognormal distribution for transaction amounts to mimic real-world financial transaction skewness, combined with randomized transaction hours, days, types, and fraud risk labels.

```
import pandas as pd
import numpy as np
np.random.seed(42)
n = 1000
data = {
    "Transaction_ID": np.arange(1, n + 1),
    "Transaction_Amount": np.round(np.random.lognormal(7, 1, n), 2),
    "Transaction_Hour": np.random.randint(0, 24, n),
    "Transaction_Day": np.random.randint(1, 31, n),
    "Transaction_Type": np.random.choice(["Online", "POS", "ATM"], n),
    "Location_Type": np.random.choice(["Local", "Domestic", "International"], n),
    "Transaction_Status": np.random.choice(["Normal", "Suspicious"], n, p=[0.90, 0.10])
}
df = pd.DataFrame(data)
```

2. Initial Preview & Record Structure

Below is a preview of the first 5 records generated by the dataset pipeline:

Tx ID	Amount (\$)	Hour	Day	Type	Location	Status
1	1,802.11	22	18	ATM	Domestic	Normal
2	955.02	11	20	ATM	International	Normal
3	2,095.80	16	17	Online	Local	Normal
4	5,029.27	7	27	Online	International	Normal
5	867.70	10	28	Online	Local	Normal

3. Data Integrity & Validation Checks

Validation scripts were run to confirm data completeness and check for duplicate transactions across the 1,000 generated entries.

1,000 TOTAL RECORDS	0 MISSING VALUES	0 DUPLICATES	7 TOTAL FIELDS
------------------------	---------------------	-----------------	-------------------

4. Statistical Distribution & Risk Analysis

Summary statistics highlight a right-skewed transaction amount distribution typical of banking fraud detection datasets.

Metric	Amount Value (\$)	Attribute Breakdown	Counts
Mean	\$1,844.90	Transaction Type (Online / ATM / POS)	346 / 335 / 319
Median (50%)	\$1,124.73	Location Type (Intl / Local / Dom)	343 / 331 / 326
Min / Max	\$42.89 / \$51,675.11	Status (Normal vs. Suspicious)	910 Normal / 90 Suspicious
Std Dev	\$2,682.53	Suspicious Rate	9.0%

5. Visualization Script & Key Findings

The transaction amounts follow a heavy-tailed log-normal distribution with an upper maximum reaching \$51,675.11.

```
import matplotlib.pyplot as plt
amount = df["Transaction_Amount"]
print("Mean:", amount.mean())
print("Median:", amount.median())
plt.figure(figsize=(8, 5))
plt.hist(amount, bins=30, color='#1e3a8a', edgecolor='white')
plt.xlabel("Transaction Amount")
plt.ylabel("Number of Transactions")
plt.title("Transaction Amount Distribution")
plt.show()
```

GitHub Repository Summary Notes: This project demonstrates end-to-end data generation, data quality assurance, categorical frequency analysis, and statistical visualization for credit card risk detection workflows.

